

Sobre el modelado
computacional del
comportamiento humano

Article

Expertise increases planning depth in human gameplay

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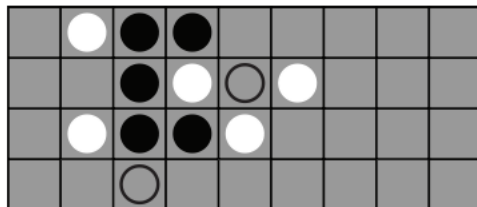
Bas van Opheusden^{1,2}✉, Ionatan Kuperwajs¹, Gianni Galbiati^{1,3}, Zahy Bnaya¹, Yunqi Li¹ & Wei Ji Ma¹

A hallmark of human intelligence is the ability to plan multiple steps into the future^{1,2}. Despite decades of research^{3–5}, it is still debated whether skilled decision-makers plan more steps ahead than novices^{6–8}. Traditionally, the study of expertise in planning has used board games such as chess, but the complexity of these games poses a barrier to quantitative estimates of planning depth. Conversely, common planning tasks in cognitive science often have a lower complexity^{9,10} and impose a ceiling for the depth to which any player can plan. Here we investigate expertise in a complex board game that offers ample opportunity for skilled players to plan deeply. We use model fitting methods to show that human behaviour can be captured using a computational cognitive model based on heuristic search. To validate this model, we predict human choices, response times and eye movements.

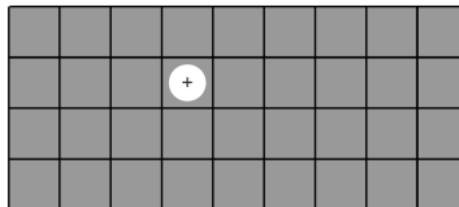
Papers Goals

- Study planning depth in humans
- Measuring it is hard because
 - Complexity often not enough to distinguish expert and novice (e.g. two-step task)
 - Tractable for modelling (not like chess)
 - Novel, simple rules and interesting to engage with
- Four-In-A-Row has all of the above

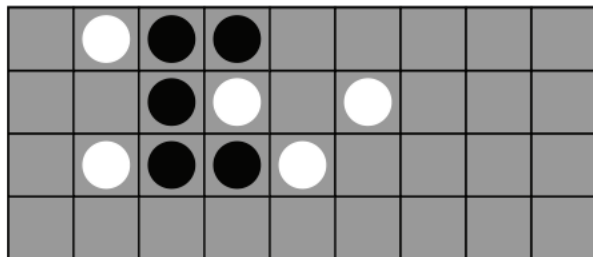
A



D



B

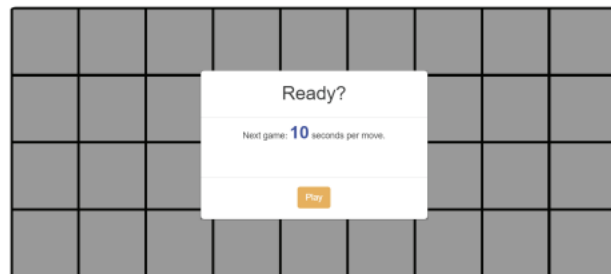


losing equal winning

C



10



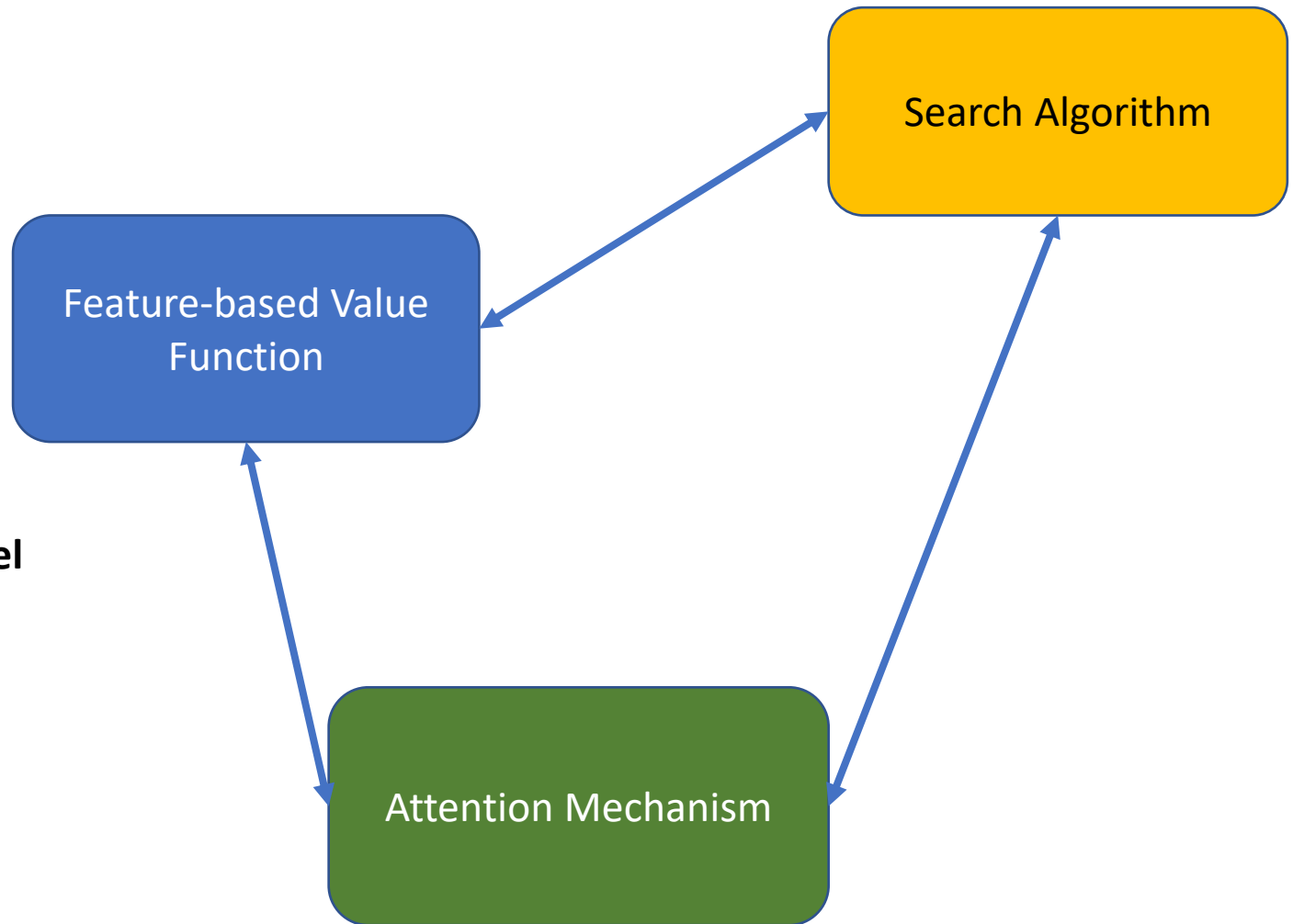
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E



Model

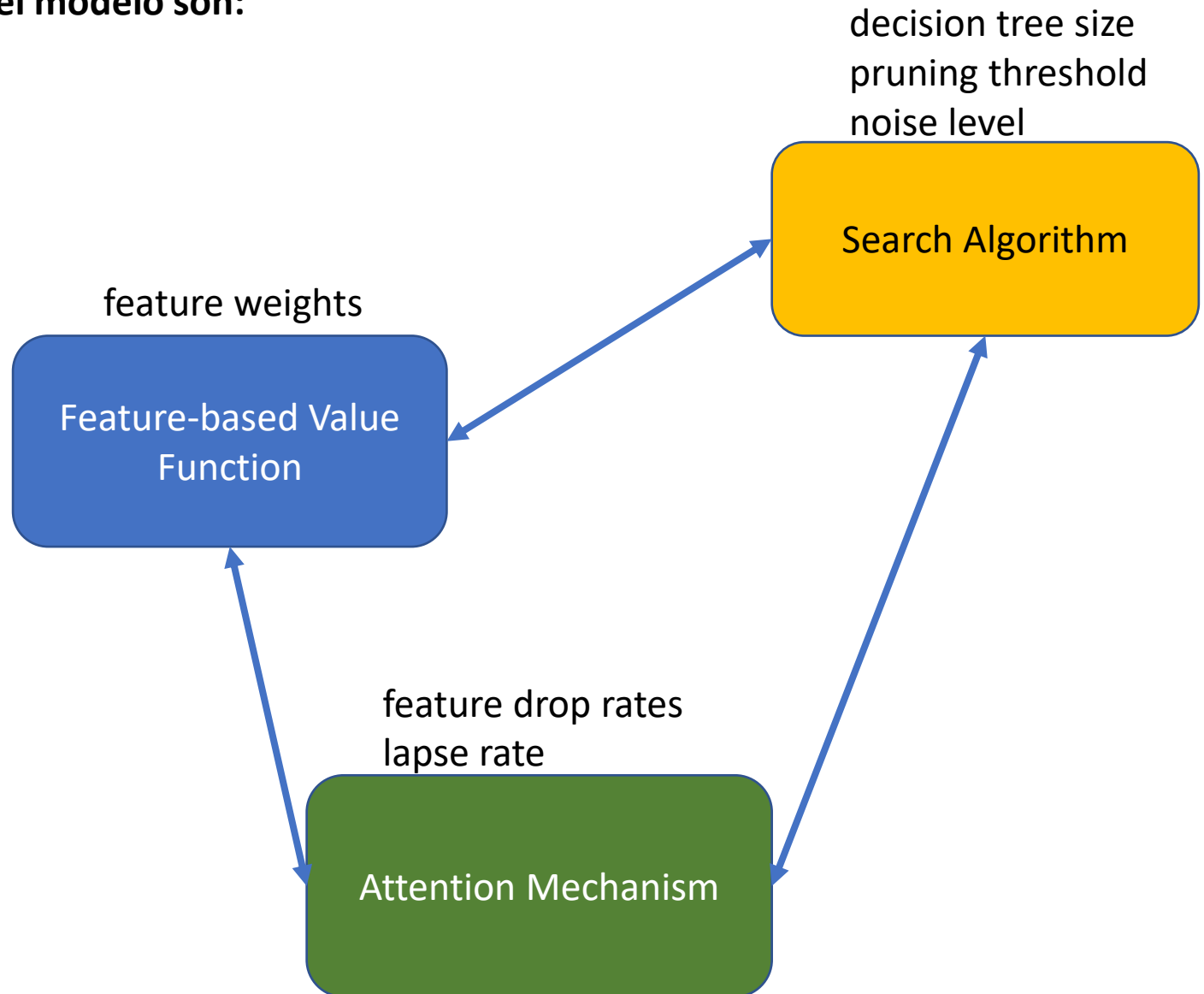
El modelo tiene tres componentes:



Los parámetros del modelo son:

feature weights
feature drop rate
labse rate
decision tree size
pruning threshold
noise level

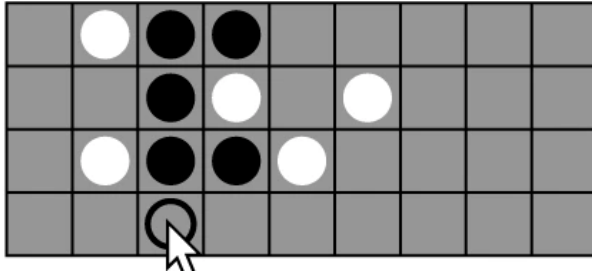
Los parámetros del modelo son:



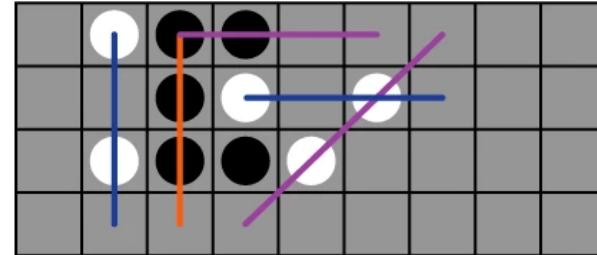
Feature-based Value
Function

Value Heuristic

a



b



- (a) Black is about to win by moving on the third square in the bottom row (open circle, mouse cursor).
- (b) Features used in the heuristic function.
- Features with identical colours are constrained to have identical weights.
- The model also includes a central tendency feature and a four-in-a-row feature.

Value Heuristic

Value function. The core of our model is a value function $V(s, \mathbf{w})$, which assigns a value to a board state s . The higher this value, the more likely the black player is to win from that state. We assume that people use value function approximation⁵⁸, and that people's value function is a weighted sum of features

$$V(s, \mathbf{w}) = \sum_{i=1}^5 w_i \phi_i(s, \text{self}) - \sum_{i=1}^5 w_i \phi_i(s, \text{opponent}) \quad (1)$$

where ϕ_i denotes the features and w_i the weights. In the following, and in the main text, we omit the dependence of $V(s, \mathbf{w})$ on $\mathbf{w}$ for brevity. The value function uses five features: centre, connected two-in-a-row, unconnected two-in-a-row, three-in-a-row and four-in-a-row. The centre feature assigns a higher value to squares near the centre of the board. The other features count how often their corresponding patterns occur on the board (horizontally, vertically or diagonally).

Whenever the model evaluates a state, the weights of features belonging to the active player are multiplied by a scaling constant C . This captures value differences between active and passive features. For example, a three-in-a-row feature signals an immediate win on the active player's move but not the opponent's (it can be blocked). We do not scale the centre feature.

Search Algorithm

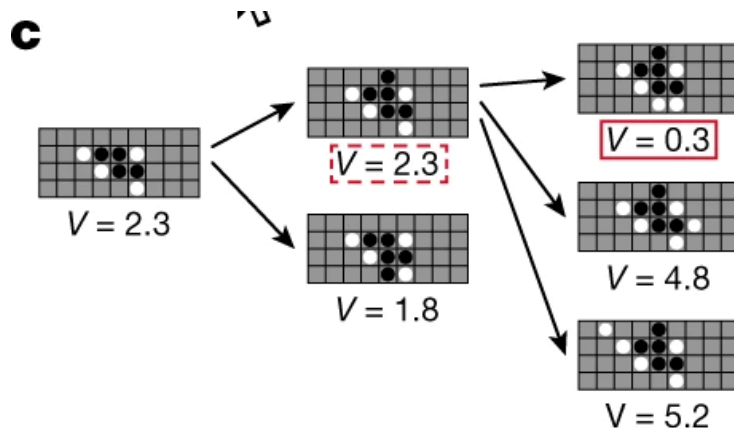
Search Algorithm

Search Algorithm

- Each iteration, the algorithm chooses a board position to explore further, evaluates the positions resulting from each legal move and prunes all moves with value below that of the best move minus a threshold.
- After each iteration, the algorithm stops with a probability γ , resulting in a geometric distribution over the total number of iterations.

Search Algorithm

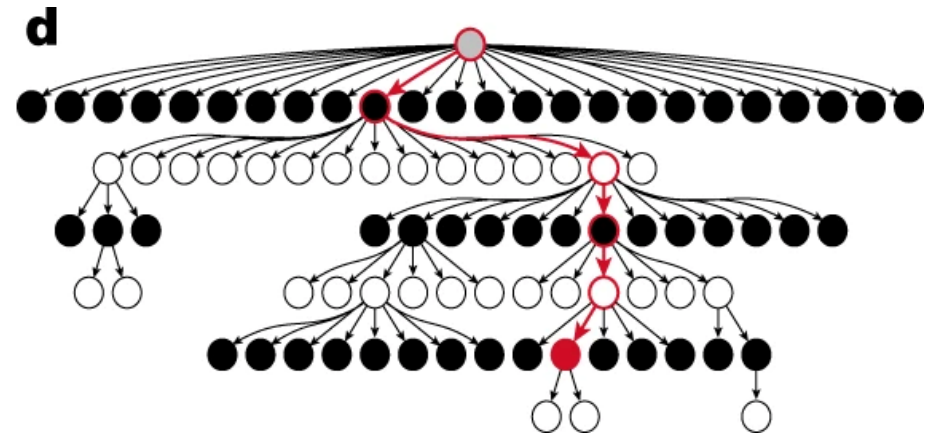
Search Algorithm



In the root position (left), black is to move.

After expanding the root node with two candidate moves for black and evaluating the resulting positions using $V(s)$, the algorithm selects the highest-value node ($V = 2.3$) on the second iteration and expands it with three candidate moves for white.

The algorithm evaluates the resulting positions, and backpropagates the lowest value ($V = 0.3$), as white is the opponent.



The decision tree built by the model with fitted parameters on an example board. The red nodes indicate the principal variation—the sequence of highest-value moves for both players. Note that different branches are evaluated to different depths.

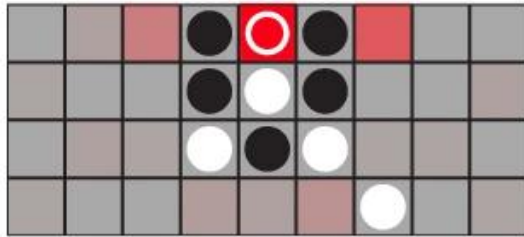
Attention Mechanism

Noise (attention)

Attention Mechanism

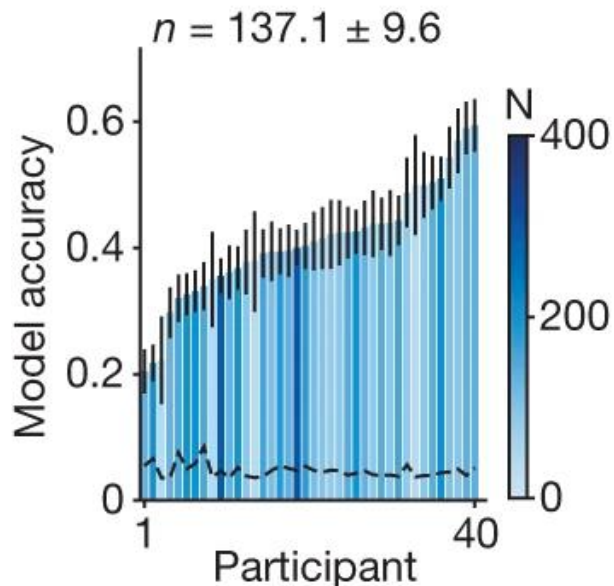
- We model selective attention by randomly dropping features (at specific locations and orientations) before constructing the decision tree, which are then omitted during the calculation of $V(s)$ anywhere in the tree.
- During the tree search, we add Gaussian noise to $V(s)$ in each node.
- a lapse rate λ for action selection (cada tanto se cofunde y no elige lo que debería).

Results

a

Human vs Human

Example board position from a human-versus-human game. The open circle indicates the move that the active player (white) chose. The red shading indicates the probability distribution of that participant's next move, as predicted by the model with parameters inferred for that participant using fivefold cross-validation.

b

Model accuracy (percentage of correctly predicted moves) for each participant in human-versus-human games, ranked from worst to best predicted. Data are mean $\pm$ s.e.m. n denotes the number of trials per participant. The dashed line represents the accuracy of a 'chance' model, which assumes that people move onto a randomly selected unoccupied square

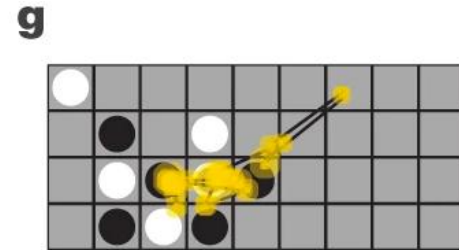
Five fold cross validation
Predicting out of sample choices

Eye Movements

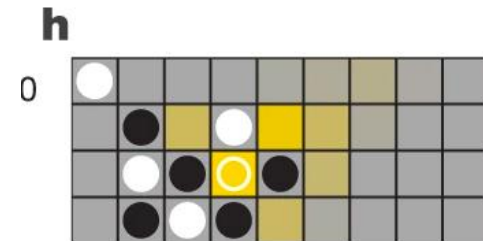
Game against computer

the distribution of squares visited by the cognitive model during its search process resembles the distribution of attention (mean correlation across participants: $\rho = 0.535 \pm 0.024$, $P < 0.001$).

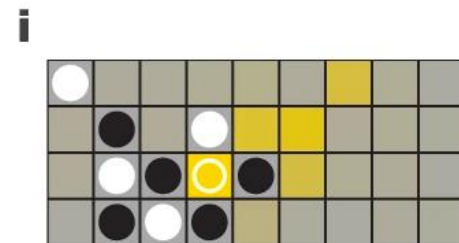
g, The trajectory of eye movements on one example trial. The black lines represent saccades and the yellow circles represent fixations. The circle area indicates the duration of fixation.



h, The estimated distribution of overt attention across unoccupied squares, obtained by convolving the eye trajectory with a Gaussian filter.

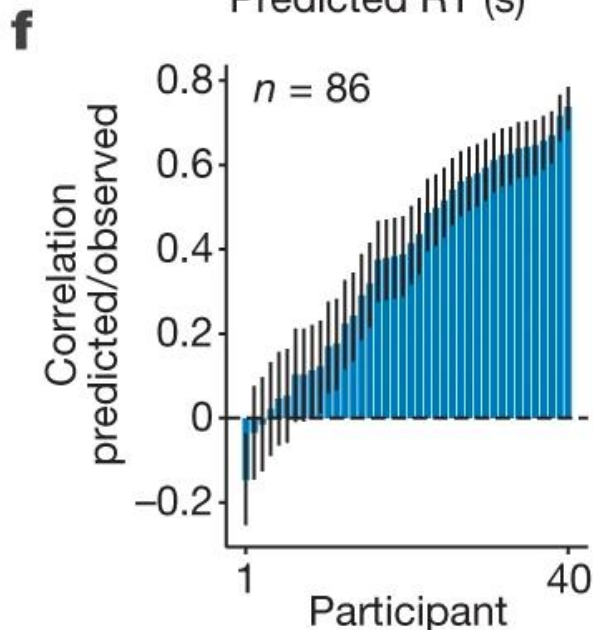
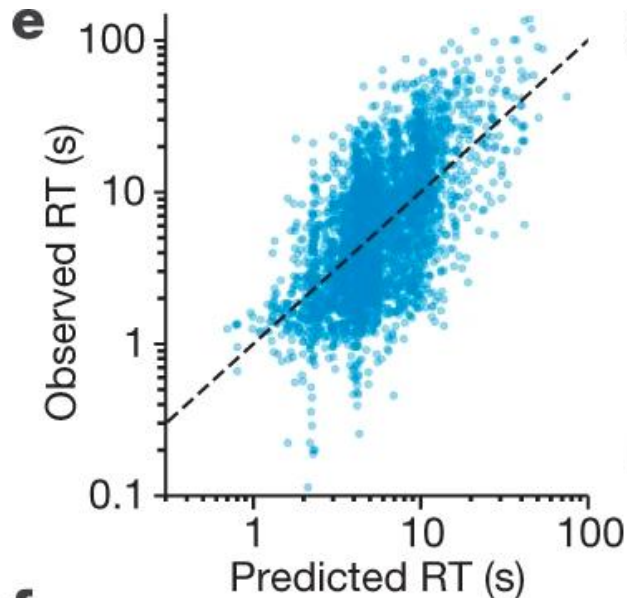


i, The distribution of squares visited by the model's search algorithm, with parameters estimated from the participant's choices.



Predicting RT

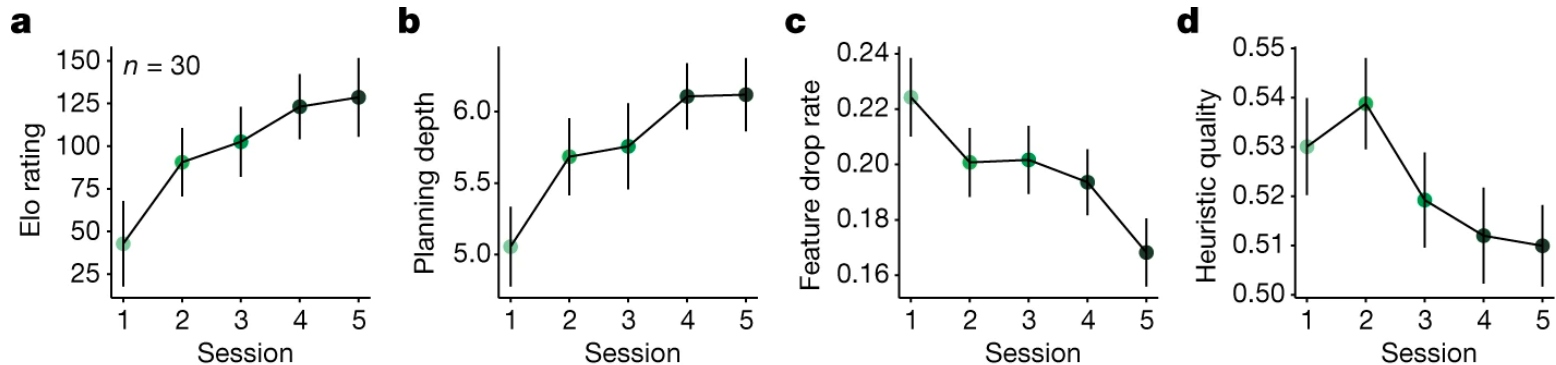
We use the decision tree built by the model on each trial as a predictor for response time.



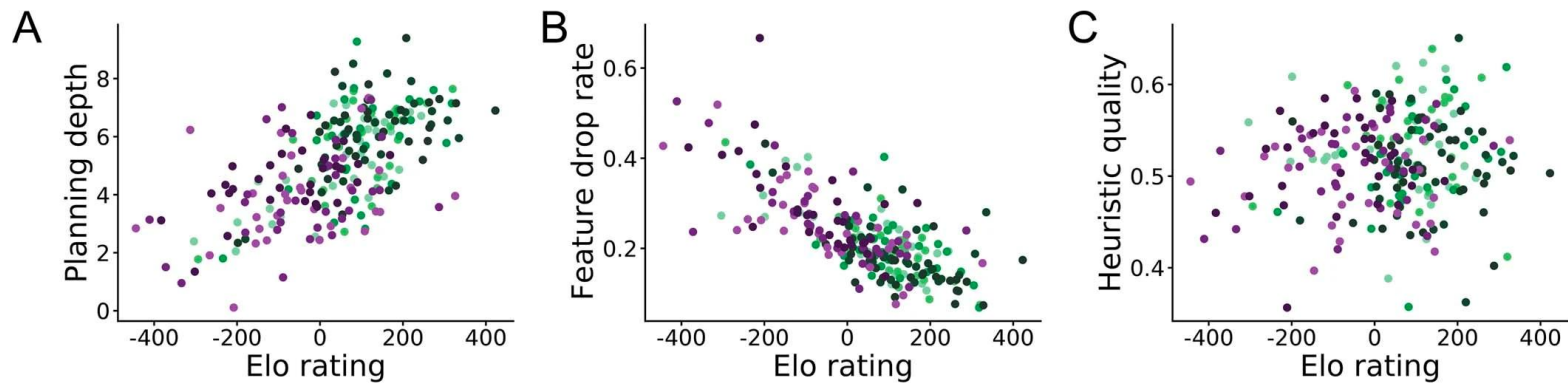
e, Predicted and observed response times (RT) across all participants in the human-versus-human data. We exclude any positions with fewer than 6 or more than 30 pieces on the board.

f, The correlation between predicted and observed response times for each participant.

- Planning depth as the length of the principal variation in the tree (the sequence in which both players always make the best moves)— an approximate measure of the number of steps that an individual thinks ahead.
- The feature drop rate is defined as the attentional lapse probability, a model parameter
- The heuristic quality as the correlation between heuristic and objective value— measuring the ‘correctness’ of the feature weights



Extended Data Fig. 8, we show that individual differences in playing strength are also correlated with planning depth and feature drop rate, and not heuristic quality



CONCLUSIÓN:

- **Stronger players plan deeper and have fewer lapses of attention.**
- **No evidence for improvements in feature weights (heuristic quality).**

Hipótesis sobre dónde se procesan en el cerebro humano los componentes del modelo

Componente del modelo

Implementación neural discutida

Feature-based Value Function

Orbitofrontal cortex

Search Algorithm

Prefrontal cortex, cingulate cortex and hippocampus

Attention Mechanism

Frontoparietal network

Idea loca: usar datos de Moravec para
decomponer planning depth entre pattern
recognition y processing speed

At the introduction:

“This literature has explained the superior performance of experts as being due to better pattern recognition and/or deeper search”

In the last paragraph previous to the discussion, under the title “Planning depth or pattern recognition”:

“experts sharpen their representation of game-relevant features, allowing for more position evaluations per unit of time and therefore deeper planning. Thus, our results are consistent with improved pattern recognition in experts, but highlight the underappreciated role of processing speed”.

Here they seem to equate “planning deep” with “processing speed”, the role of pattern recognition being to accelerate the processing speed. But I think this is misleading and that it is important to understand if we become experts by performing more conscious computation per unit time or by pattern recognition, which are not the same thing.

I can use the RT for 2x2 and 3x2 to estimate the expected RT for 23x2 if the time to combine the two previous results were negligible: the calculation of 23x2 would take the sum of the RTs for 2x2 and 3x2.

Any complex mental calculation involves evoking the multiplications and/or addition tables, i.e., evoking one-digits multiplications and/or additions (1x1, 1x2,..., 9x8, 9x9 and 1+1, 1+2,..., 9+8, 9+9). For instance, to calculate 16x2 I need to evoke the result for 1x2, then the result for 6x2 (mental calculation is performed from left to right, contrary to pen and pencil math), and, lastly, the result for 2+1 (to add the tens). But, after some training, I may just remember that 16x2 equals 32, without the need to perform any computation.

So, mental calculation can improve by two distinct mechanisms: processing speed and pattern recognition

